Multiplying and Dividing Decimals by 10, 100, ...

When multiplying by 10, the number is increased by moving the decimal point to the right:

Example: $25.12 \times 10 = 251.2$

and $25.12 \times 100 = 2512$

Similarly when dividing by 10, 100 etc., the number is decreased by moving the decimal point to the left:

Example: $432.1 \div 10 = 43.21$

and $432.1 \div 100 = 4.321$

Multiplying and Dividing Decimals

When multiplying two decimals without a calculator, first ignore the point and write as whole numbers and then multiply. Then insert the decimal point by counting the digits after the decimal points in the numbers being multiplied and having the same number of digits after the decimal point in the result.

$1.5 \times 0.6 \quad : \quad 15 \times 6 = 90$

$\Rightarrow 1.5 \times 0.6 = 0.90$

When dividing by a decimal change the divisor into a whole number by multiplying the numerator and denominator by 10 or 100 or by making an equivalent fraction without a decimal in the denominator.

(i) $\frac{1.2}{0.4} = \frac{12}{4} = 3.0$

(ii) $\frac{0.225}{0.15} = \frac{22.5}{15} = 1.5$

Exercise 1D

(The following exercise should be done without the use of a calculator.)

1. Write down the value of the digit 4 in each of the following numbers:

(i) 2.4 (ii) 2.47 (iii) 16.24 (iv) 1.042 (v) 8.254

2. Express each of the following decimals as a fraction in its lowest terms:

(i) 0.4 (ii) 0.5 (iii) 0.75 (iv) 0.25 (v) 0.45
(vi) 0.15 (vii) 0.05 (viii) 0.125 (ix) 0.375 (x) 0.625

3. Write the following fractions as decimals:

(i) $\frac{1}{2}$ (ii) $\frac{1}{4}$ (iii) $\frac{3}{5}$ (iv) $\frac{5}{8}$ (v) $\frac{8}{10}$

4. Write down the answer to each of these:

(i) 1.6×10 (ii) 1.64×10 (iii) 2.34×100
(iv) 0.74×100 (v) 1.28×1000 (vi) 12.07×100

5. Write down the value of each of these:

(i) $346 \div 10$ (ii) $1.84 \div 10$ (iii) $12.3 \div 10$
(iv) $546 \div 100$ (v) $23.4 \div 100$ (vi) $128.4 \div 1000$

6. Find the value of each of the following:

(i) 4.6×8 (ii) 12.34×4 (iii) 1.67×8 (iv) 1.2×0.6
(v) 3.8×0.7 (vi) 4.64×0.8 (vii) 1.21×0.6 (viii) 1.5×0.08

7. Express each of the following as a whole number or as a fraction in its simplest form:

(i) $\frac{3.6}{0.9}$ (ii) $\frac{2.8}{0.7}$ (iii) $\frac{0.24}{0.04}$ (iv) $\frac{0.8}{4.8}$ (v) $\frac{0.56}{0.8}$

8. Find the value of:

(i) $34.76 + 2.15 - 311.4$ (ii) $25.12 - 3.45 - 16.21$ (iii) $47.12 \times 13.5 - 27.63$